

W7 - 1.6 Determine a Quadratic Equation Given its Roots

MCR3U

1) Find the equation of a quadratic function that has the given x-intercepts and contains the given point. Express each function in factored form.

a) -3 and 5, point (4, -3)

b) -4 and 7, point (-3, -12)

2) Express each function from #1 in standard form.

a)

b)

3) Find the equation of a quadratic function that has the given x-intercepts and contains the given point. Express each function in standard form.

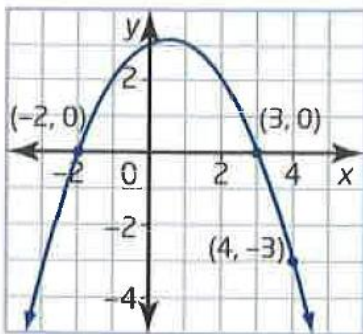
a) $1 \pm \sqrt{11}$, point (4, -6)

b) $-2 \pm \sqrt{7}$, point (1, 2)

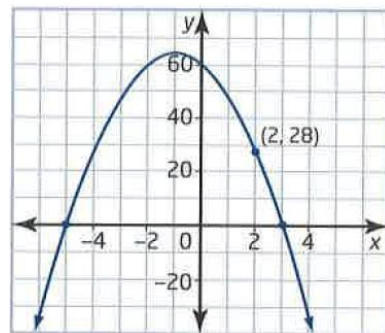
4) Find the equation of the quadratic function that has zeros at 3 and -1 and contains the point (1, -2). Express the function in vertex form.

5) Determine the equation in standard form for each quadratic function shown.

a)



b)



6) An arch of a highway overpass is in the shape of a parabola. The arch spans a distance of 12 meters from one side of the road to the other. The height of the arch is 8 meters at a horizontal distance of 2 meters from each side of the arch.

a) Sketch the quadratic function if the vertex of the parabola is on the y-axis and the road is along the x-axis.

b) Use this information to determine the function that models the arch.

c) Find the maximum height of the arch to the nearest tenth of a meter.